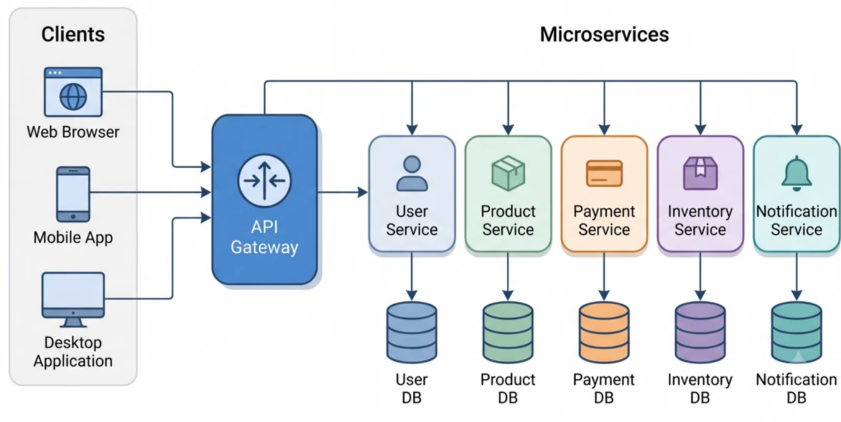
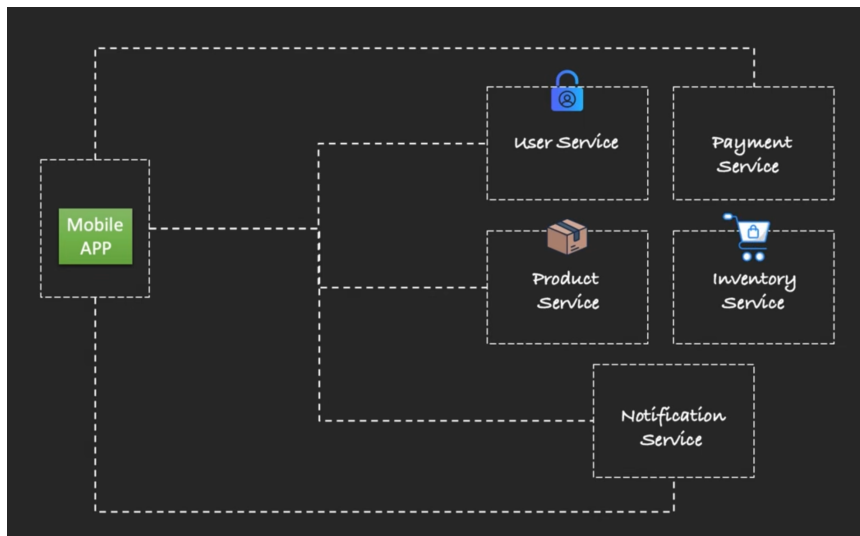


## API Gateway

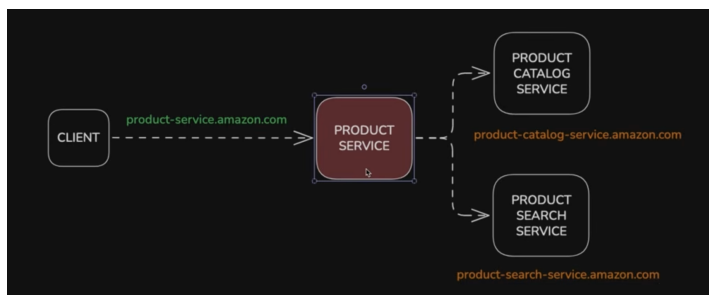


## Without API Gateway



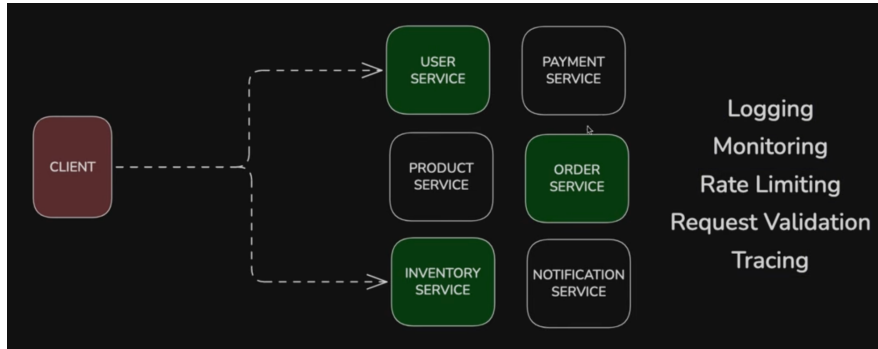
## Problem1: CLIENT NEED TO KNOW EVERYTHING

Client needs to know everything about your server. In client application we need to configure all the micro services. Tight coupling. What if it is keep on growing.

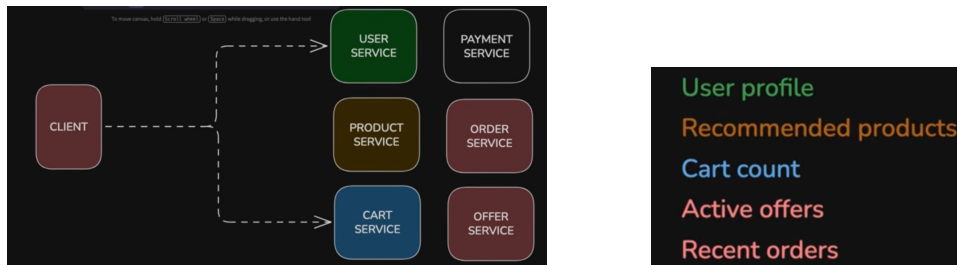


## Problem 2: DUPLICATE CROSS CUTTING LOGIC

Each service should maintain cross cutting logic. For example, security need to implement for every service. Common logic we need to duplicate in every service.



## Problem 3: TO MANY NETWORK CALLS:



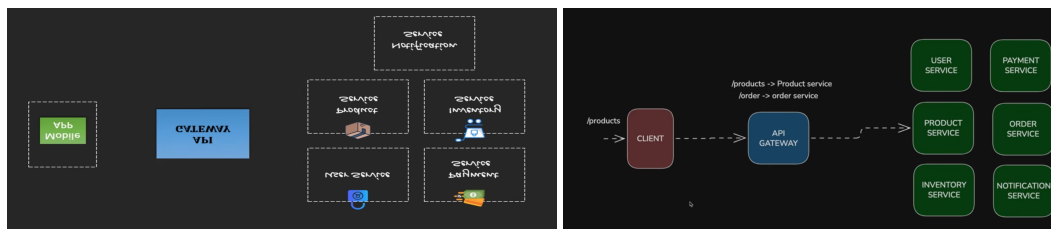
Client needs to call multiple service to display page.

latency + failures + timeout we can notice in slow network.

## SOLUTION – API GATEWAY.

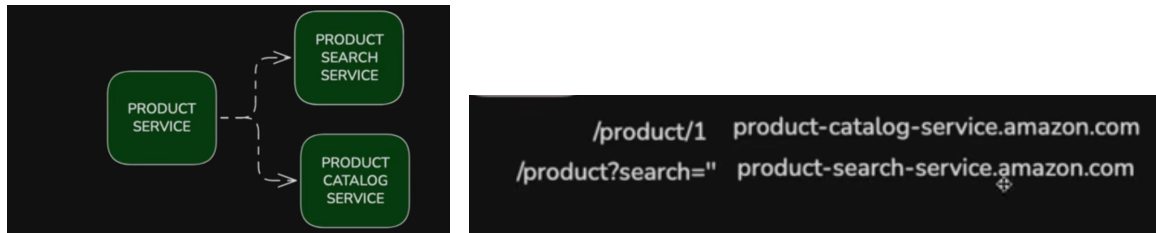
**SOLUTION 1:** Single entry point and route the request to corresponding service.

Client need not to know about how many micro services behind the scenes are working.



Mapping will be available the API gateway

We will update only API gateway not client application. New mappings only at API Gateway.



**SOLUTION 2:** Instead of having secondary logic in every micro service, we can abstract out that logic to API gateway.

**SOLUTION 3:** API Gateway makes all the calls required for a page, combine them and return. Request and Response Transformation logic in API gateway.

**SOLUTION 4:** Cache logic for same request